Experiment 5(a): Simulation and Performance Evaluation of RFID/NFC-Based Device Identification and IoE Data Integration using Node-RED and MQTT

Objectives

1. To simulate RFID/NFC-based device identification using Node-RED.
2. To transmit device IDs using MQTT.
3. To process and validate device data using Node-RED.
4. To visualize device information on a dashboard.

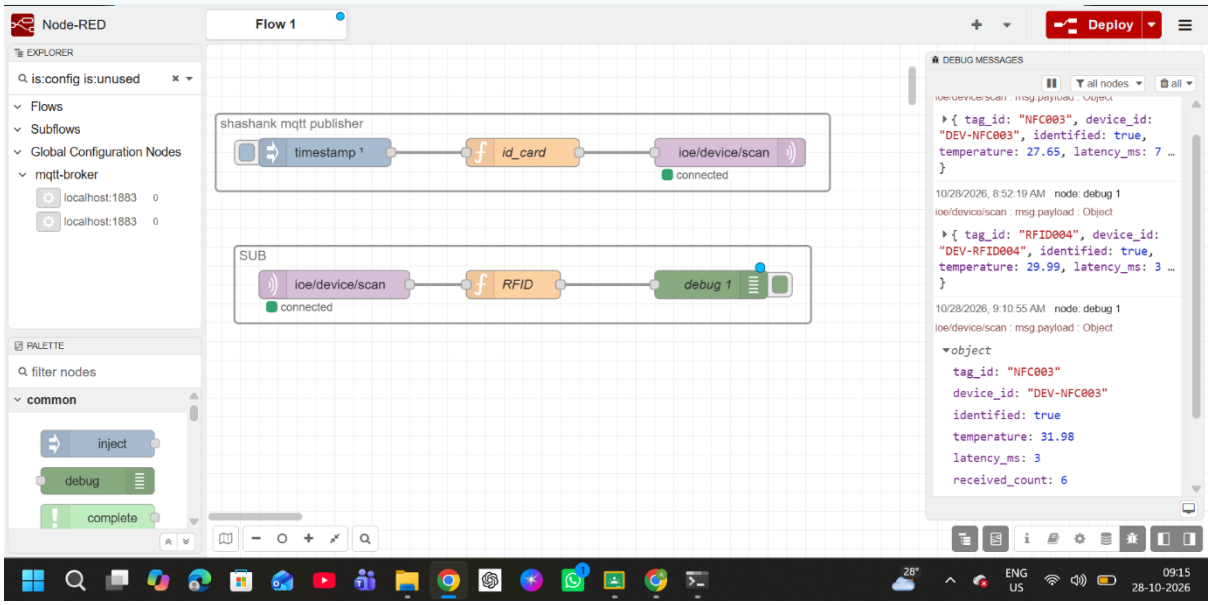
Software Required

- Node-RED
- MQTT broker (Mosquitto or Node-RED MQTT nodes)
- Node-RED Dashboard

Procedure

1. Open Node-RED and create a new flow.
2. Configure an Inject node to simulate RFID/NFC tag IDs.
3. Connect a Function node to process device IDs and generate sample sensor data.
4. Configure an MQTT Output node to publish data to ioe/device/id.
5. Use an MQTT Input node to receive the published messages.
6. Connect Debug nodes to verify the received data.
7. Display device IDs and sensor readings on the Node-RED dashboard.

Output:



Output Table

S.No.	Parameter	Expected Output
1	Device ID	Valid simulated ID
2	MQTT topic	ioe/device/id
3	Message transmission	Successful
4	Data processing	Valid device data
5	Dashboard	Device details displayed

Result

RFID/NFC-based device identification and IoE data integration were simulated successfully using Node-RED and MQTT.